

Fluid mechanics holistic

idiabix section one

The fluid property, due to which, mercury does not wet the glass is

- A. surface tension
- B. viscosity
- C. cohesion
- D. adhesion

Answer: Option A

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

Discuss

2. Laminar flow of a Newtonian fluid ceases to exist, when the Reynolds number exceeds

- A. 4000
- B. 2100
- C. 1500
- D. 3000

Answer: Option B

Explanation:

No answer description available for this

question. Let us discuss .

[View Answer](#)

[Discuss](#)

3. When the momentum of one fluid is used for moving another fluid, such a device is called a/an

A. jet pump

B. blower

C. acid egg

D. none of these

Answer: Option A

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

4. The normal stress is the same in all directions at a point in a fluid, when the fluid is

A. non-viscous.

B. incompressible.

C. both (a) and (b).

D. having no motion of one fluid layer relative to the other.

Answer: Option D

Explanation:

No answer description available for this

question. Let us discuss .

[View Answer](#)

[Discuss](#)

5. Head developed by a centrifugal pump depends on its

- A. speed
- B. impeller diameter
- C. both (a) and (b)
- D. neither (a) nor (b)

Answer: Option C

6. Hydraulic mean depth (D_m) for a circular pipe of diameter ' D ' flowing full is $0.25 D$. For a circular channel, at $D_m = 0.3 D$, gives the condition for the maximum

- A. flow rate
- B. mean velocity
- C. both 'a' & 'b'
- D. neither 'a' nor 'b'

Answer: Option B

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

7. The head loss in turbulent flow in a pipe varies

- A. as velocity
- B. as (velocity)²
- C. inversely as the square of diameter
- D. inversely as the velocity

Answer: Option B

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

8. Most commonly used joint in the underground pipe lines is the

- A. sleevejoint
- B. coupling
- C. flange
- D. expansion joint

Answer: Option A

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

9. Schedule number of a pipe, which is a measure of its wall thickness, is given by

- A. $1000 \sqrt{P/S}$
- B. $100 \sqrt{P/S}$

C. 1000 S/P'

D. 10000 P'/S

Answer: Option A

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

10. The net positive suction head (NPSH) of a centrifugal pump is defined as the sum of the velocity head and the pressure head at the

A. discharge.

B. suction.

C. suction minus vapor pressure of the liquid at suction temperature.

D. discharge minus vapor pressure of the liquid at the discharge temperature.

Answer: Option C

11. With increasing flow rate, the hydraulic efficiency of a centrifugal pump

A. monotonically decreases.

B. decreases and then increases.

C. remains constant.

D. increases and then decreases.

Answer: Option A

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

12. The velocity profile for turbulent flow through a closed conduit is

- A. logarithmic
- B. parabolic
- C. hyperbolic
- D. linear

Answer: Option A

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

13. In case of hydraulically smooth pipe, the resistance to flow depends only on the Reynolds number, whereas for a hydraulically rough pipe, the resistance to flow is governed by the relative roughness. Two pipes are said to have the same hydraulic roughness, when they have equal values of

- A. relative roughness.
- B. absolute roughness.
- C. friction co-efficient for flows at equal

Reynold number.

D. all (a), (b) & (c).

Answer: Option C

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

14. Water hammer in a pipeline results from the

A. bursting of pipelines due to closure by a valve.

B. rapid pressure change due to a rapid change in the rate of flow.

C. pressure increase due to closure of a valve resulting in decrease in rate of flow.

D. none of these.

Answer: Option B

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

15. Molten soap mass is transported by a _____ pump.

- A. diaphragm
- B. reciprocating
- C. gear
- D. centrifugal

Answer: Option D

16. Which of the following denotes the effect of compressibility in fluid flow ?

- A. Weber number
- B. Mach number
- C. Euler number
- D. Reynolds number

Answer: Option B

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

17. Nominal size of the discharge pipe of a pump is usually _____ the nominal size of the inlet pipe.

- A. smaller than
- B. larger than
- C. same as
- D. twice

Answer: Option A

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

18. The terminal velocity of a small sphere settling in a viscous fluid varies as the

- A. first power of its diameter.
- B. inverse of the fluid viscosity.
- C. inverse square of the diameter.
- D. square of the difference in specific weights of solid & fluid.

Answer: Option B

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

19. If the discharge of a centrifugal pump is throttled, then its suction lift

- A. increases
- B. decreases
- C. remains unchanged
- D. data insufficient to predict

Answer: Option A

21. At high Reynolds number

- A. inertial forces control and viscous

forces are unimportant.

B. viscous forces predominate.

C. inertial forces are unimportant and viscous forces control.

D. none of these.

Answer: Option A

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

22. Power loss in an orificemeter is _____ that in a venturimeter.

A. less than

B. same as

C. more than

D. data insufficient, cannot be predicted

Answer: Option C

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

23. The terminal velocity of a particle moving through a fluid varies as $\rho \mu^n$. What is the value of n for Newton's law regime ?

A. 0.5

B. 1

C. 1.5

D. 3

Answer: Option A

26. A particle A of diameter 10 microns settles in an oil of specific gravity 0.9 and viscosity 10 poise under Stoke's law. A particle B with diameter 20 microns settling in the same oil will have a settling velocity

A. same as that of A.

B. one fourth as that of A.

C. twice as that of A.

D. four times as that of A.

Answer: Option B

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

27. The pitot static tube does not measure the _____ pressure.

A. static

B. total

C. difference in static & dynamic

D. all (a), (b) and (c)

Answer: Option D

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

28. The fluid in which the shearing stress within it is proportional to the velocity gradient across the sheared section, is called a _____ fluid.

A. Bingham

B. Newtonian

C. perfect

D. none of these

Answer: Option B

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

29. In fluid flow, cavitation is caused, if the

A. fluid velocity decreases to zero.

B. total energy decreases.

C. both (a) and (b).

D. flow pressure approaches its vapor pressure at the prevailing temperature.

Answer: Option D

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

30. The ratio of the depth of flow to the hydraulic radius for the most economical trapezoidal section, in open channel flow is

- A. 0.5
- B. 1
- C. 1.5
- D. 2

Answer: Option D

31. The drag co-efficient for a bacterium moving in water at 1 mm/s, will be of the following order of magnitude (assume size of the bacterium to be 1 micron and kinematic viscosity of water to be $10^{-6} \text{ m}^2/\text{s}$).

- A. 24000
- B. 24
- C. 0.24
- D. 0.44

Answer: Option D

Explanation:

No answer description available for this

question. Let us discuss .

[View Answer](#)

[Discuss](#)

32. For Laminar flow through a packed bed, the pressure drop is proportional to (V_s is the superficial liquid velocity and D_p is the particle diameter)

A. V_s / D_p^2

B. V_s^2 / D_p^2

C. V_s^2 / D_p^3

D. V_s / D_p^3

Answer: Option A

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

33. The maximum delivery pressure of compressors can be upto _____ atmospheres.

A. 10

B. 100

C. 250

D. 1000

Answer: Option D

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

34. Dimension of surface tension is (where, F = force, L = length)

- A. FL^{-1}
- B. $F^{-1} \cdot L$
- C. $F \cdot L^{-2}$
- D. $F^{-2} \cdot L$

Answer: Option A

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

35. Which of the following is not a dimension-less parameter ?

- A. Euler number
- B. Specific gravity
- C. Fanning friction factor
- D. None of these

Answer: Option D

36. Co-efficient of velocity is _____ the coefficient of discharge.

- A. less than

- B. more than
- C. equal to
- D. not related to

[View Answer](#)

[Discuss](#)

37. All pipes of a particular nominal size have the same

- A. inside diameter
- B. outside diameter
- C. thickness
- D. none of these

[View Answer](#)

[Discuss](#)

38. For the transfer of solution of thick slurry, the pump used is a _____ pump.

- A. reciprocating
- B. gear
- C. diaphragm
- D. centrifugal

Answer: Option C

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

39. Which of the following is not concerned with

the fluid-particle interaction ?

- A. Drag co-efficient
- B. Froude number
- C. Galileo number
- D. Weber number

Answer: Option D

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

40. Maximum theoretical suction lift for water at 15°C by a centrifugal pump is 34 ft. The same for water at 90°C will be _____ ft.

- A. 40
- B. 34
- C. 8
- D. 37

Answer: Option C

41. Pick out the wrong statement.

- A. Momentum transfer in laminar flow results from velocity gradient.
- B. A fluid in equilibrium is not free from shear stress.
- C. The viscosity of a non-Newtonian fluid is a function of temperature only.

D. both (b) and (c)

Answer: Option D

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

42. Pressure drag does not depend upon the

- A. roughness of surface of the body.
- B. pressure of main flow only.
- C. length of the body in flow direction.
- D. all (a), (b) and (c)

Answer: Option D

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

43. Dimension of absolute viscosity is

- A. $ML^{-1} T^{-1}$
- B. MLT^{-1}
- C. $ML^{-1} T$
- D. MLT

Answer: Option A

Explanation:

No answer description available for this

question. Let us discuss .

[View Answer](#)

[Discuss](#)

44. _____ pumps are axial flow pumps.

- A. Turbine
- B. Propeller
- C. Diffuser
- D. none of these

Answer: Option B

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

45. Brownian movement is prominent in the particle size range of _____ microns in case of settling of a particle in a fluid.

- A. 2 to 3
- B. 0.01 to 0.10
- C. 200 to 300
- D. 100 to 1000

Answer: Option A

46. For a particle settling in water at its terminal settling velocity, which of the following is true ?

- A. Buoyancy = weight + drag

B. Weight = buoyancy + drag

C. Drag = buoyancy + weight

D. Drag = weight

Answer: Option B

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

47. A Bingham fluid of viscosity $\mu = 10 \text{ Pa.s}$ and yield stress, $\tau_0 = 10 \text{ KPa}$, is shared between flat parallel plates separated by a distance of 10^{-3} m . The top plate is moving with a velocity of 1 m/s . The shear stress on the plate is

A. 10 KPa

B. 20 KPa

C. 30 KPa

D. 40 KPa

Answer: Option B

Explanation:

No answer description available for this question. Let us discuss .

[View Answer](#)

[Discuss](#)

48. Pick out the correct statement.

- A. A forced vortex occurs when fluid rotates as a solid about an axis.
- B. In laminar flow, Newton's law of viscosity does not apply.
- C. A free vortex occurs, when fluid rotates as a solid.
- D. In turbulent flow, there are neither cross-currents nor eddies.

Answer: Option A

50. The resistance wire used in a hot wire anemometer for conducting electrical current is made of

- [A]. copper
- [B]. tungsten
- [C]. chromium
- [D]. aluminium

Answer: Option B

Explanation:

No answer description available for this question.

Workspace Report

Hardik said: (Dec 8, 2014)

Because tungsten has a high melting point.

Nagaraju said: (Dec 22, 2016)

Actually, it is chromium. Reference :

Ramprasad book.

Mohammed said: (Jun 12, 2017)

It should be chromium.

.....End,.....section one